

Complement

Radix

$\frac{10}{2}$ $\swarrow$ 10
9

2^s comp.

2^s comp.

$\swarrow$ 0
1

Complement

Radix

10
2
10
9
0

111011
000100

^s
2 comp. →

^s
2 comp.

111011

000100 2^s comp.

000100

↓

000101

✓ 11

00

Boolean Algebra

$$2^2 = 4$$

Logic operation Truth Table

AND

$$x \cdot y = \begin{cases} 0 \\ 1 \end{cases}$$

OR

$$x + y = \begin{cases} 0 \\ 1 \end{cases}$$

NOT

$$\begin{matrix} 0 \rightarrow 1 \\ 1 \rightarrow 0 \end{matrix}$$

x	y	xy
0	0	0
0	1	0
1	0	0
1	1	1

$$\begin{array}{r} 111011 \\ \hline 000100 \end{array}$$

$$111011$$

$$000101 \cdot 2^5 \text{ Comp.}$$

Boolean Algebra

$$2^2 = 4$$

$$\begin{array}{r} 111011 \\ \hline 000100 \end{array}$$

Logic operation Truth Table

NAND

X	Y	AND xy	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1		
0	1	0	1	1	0		
1	0	0	1	1	0		
1	1	1	1	0	1		

Boolean Algebra

$$2^2 = 4$$

Logic Gates

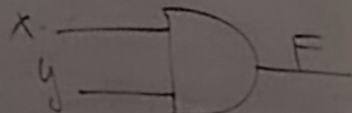
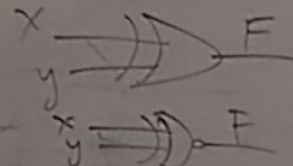
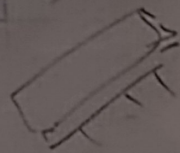
Logic operation

AND

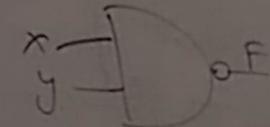
Truth Table

X	Y	AND xy	OR	AND	NOR	XOR	XNOR
0	0	0	0	1	1		
0	1	0	1	1	0		
1	0	0	1	1	0		
1	1	1	1	0	0		

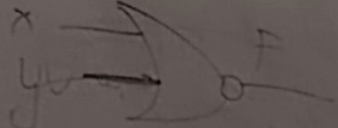
1 1 1 0 1 1
0 0 0 1 0 0



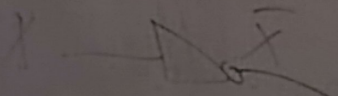
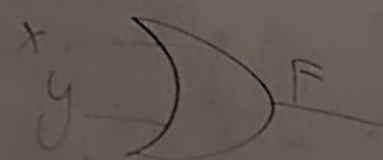
NAND



NOR



OR



$$X \neq 0 \longrightarrow X = 1$$

$$X \neq 1 \longrightarrow X = 0$$

$$X + 0 \longrightarrow X$$

$$X \cdot 0 \longrightarrow 0$$

$$X + 1 \longrightarrow 1$$

$$X \cdot 1 \longrightarrow X$$

$$X + X \longrightarrow X$$

$$X \cdot X \longrightarrow X$$

$$X + \overline{X} \longrightarrow 1$$

$$X \cdot \overline{X} \longrightarrow 0$$

$$X + y \longrightarrow y + X$$

$$X \cdot y \longrightarrow y \cdot X$$

$$X + Xy \longrightarrow X(1+y) \longrightarrow X$$

$$X + \overline{X}y \longrightarrow X + y$$

$$\overline{\overline{X}} \longrightarrow X$$

Boolean Algebra

$$2^2 = 4$$

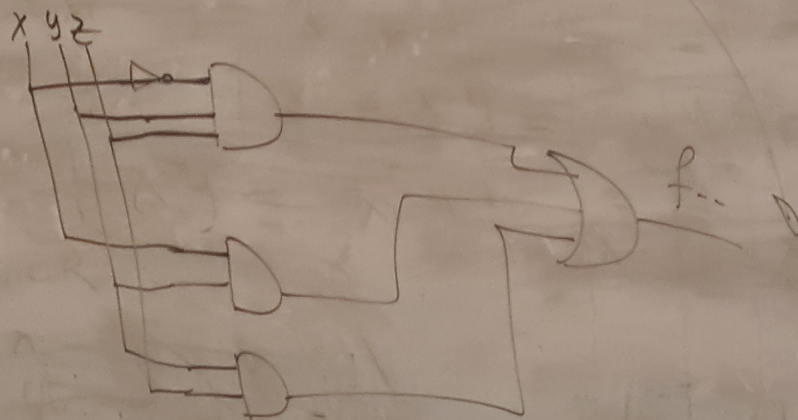
De Morgan

$$1. \overline{x + y} = \bar{x} \cdot \bar{y} \dots \dots$$

$$2. \overline{xy} = \bar{x} + \bar{y}$$

Boolean Algebra $2^2 = 4$

$$f(a,b,c) = \bar{x}yz + xy + yz$$



Q1.

Prove the following:

$$\overline{X} + XY = \overline{X} + Y$$

Q2.

Find the logic circuit for:

$$\overline{A}BC + A\overline{B}C + \overline{A}B + ABC$$